Ahmedabad University

Safety-Enhanced UAV Path Planning Using Spherical Vector-Based Particle Swarm Optimization

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Safety-Enhanced UAV Path Planning Using SPSO

Application domain

- Autonomous UAV navigation
- Terrain-aware flight planning
- Obstacle avoidance

Real-World Motivation

- Disaster monitoring
- Surveillance missions
- Mapping and terrain inspection
- Package delivery

Motivation

- UAVs must travel efficiently
- Need to avoid obstacles and dangerous areas
- Optimization algorithms help compute safe paths

Problem Statement

Goal

Need to find an optimal UAV path between:
Start Point to the End point

While satisfying

- Obstacle avoidance
- Terrain elevation constraints
- Minimum path length
- Safe flight altitude

Challenges

- Complex terrain environment
- Multiple obstacles
- Large search space

Optimization Approach

Algorithm used

Particle Swarm Optimization (PSO)

Why used PSO

- Population-based optimization
- Inspired by swarm intelligence
- Efficient in solving nonlinear optimization problems

Enhancement

- Spherical Vector Representation
- Used for smoother trajectory generation.

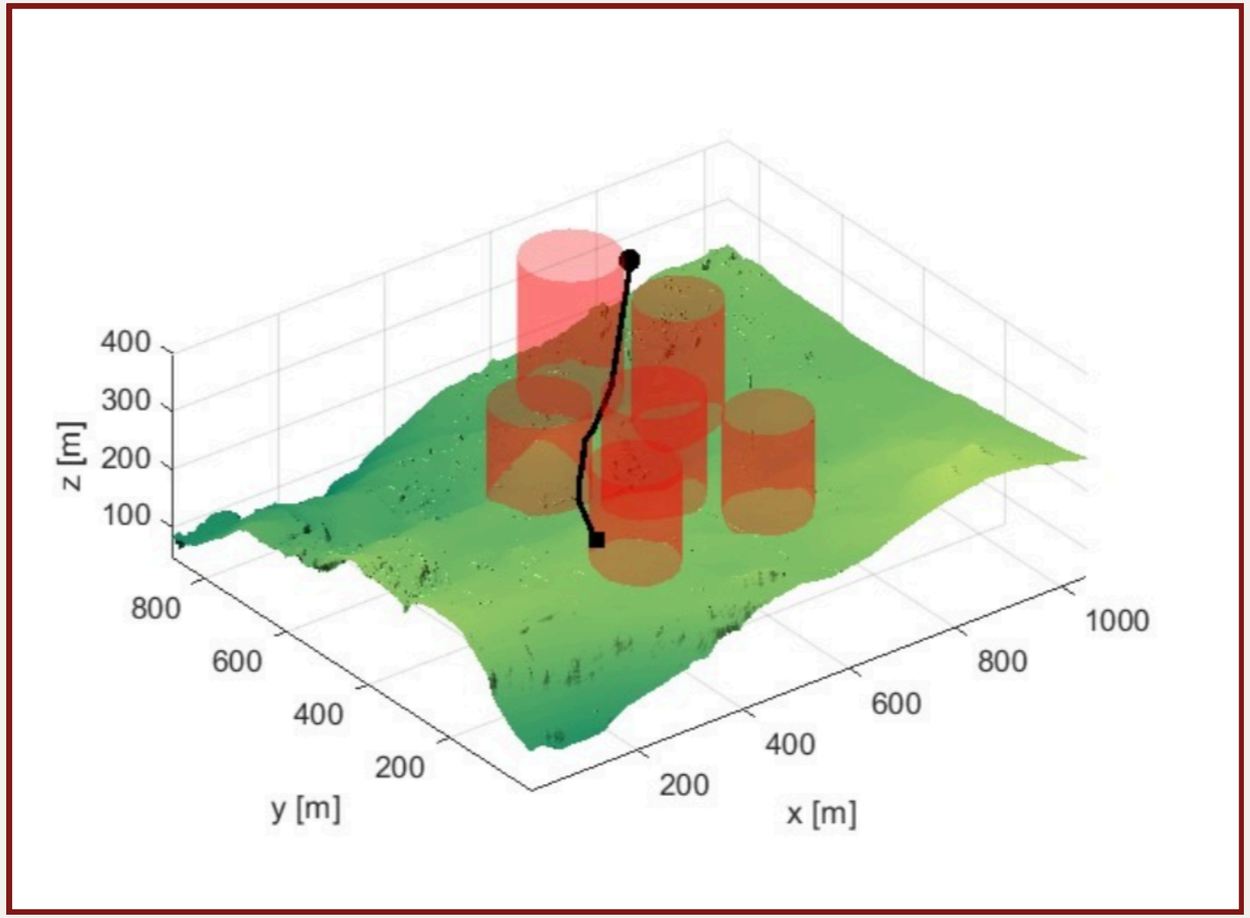
System Model

Environment consists of:

- 3D terrain map
- Cylindrical obstacles
- Start position
- Target position

Path Representation

- Coordinates defined using:
- Distance (r)
- Elevation angle (ψ)
- Direction angle (φ)



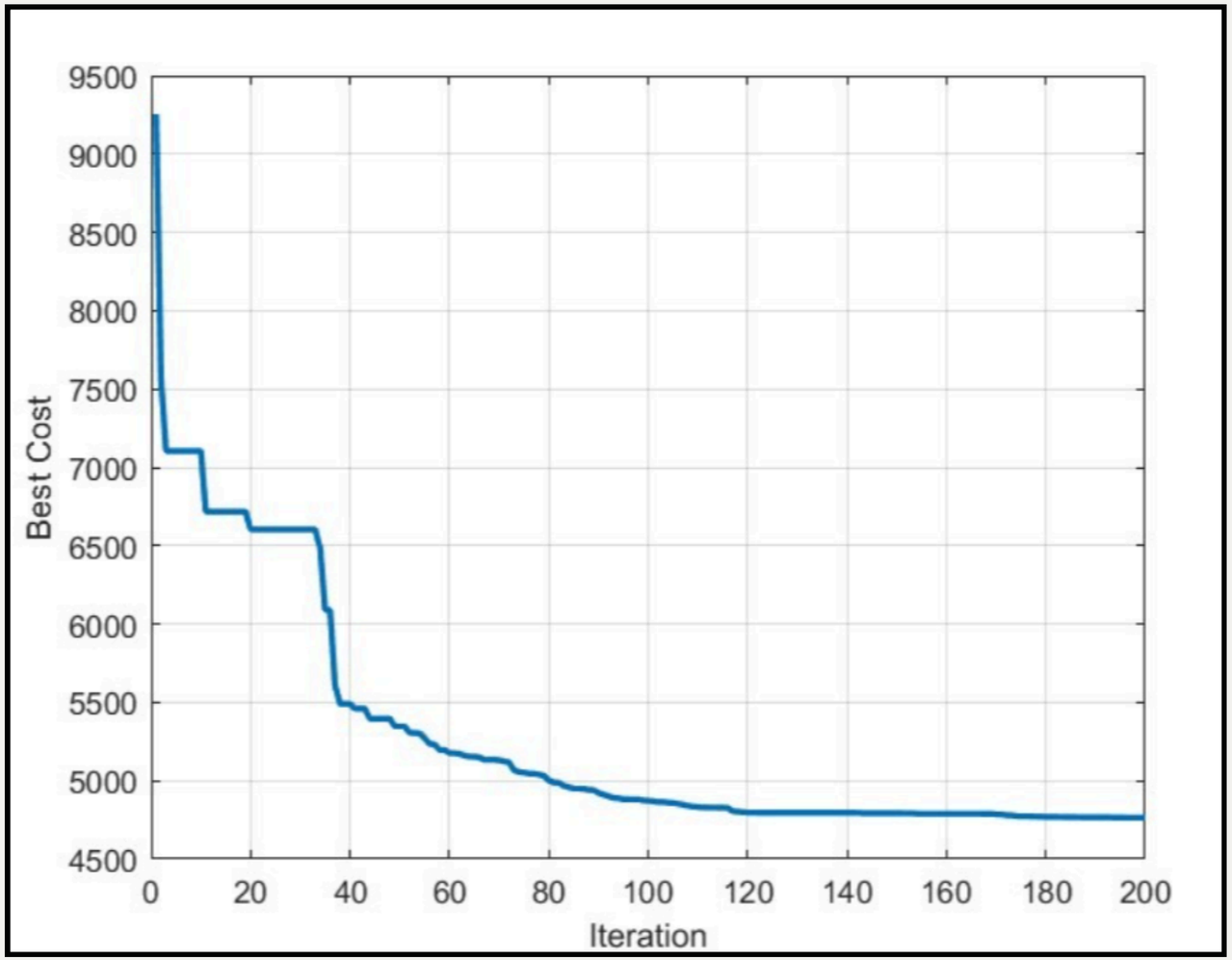
Cost Function

Components include:

- Path length
- Collision penalty
- Altitude violation penalty
- Smoothness of trajectory

Goal:

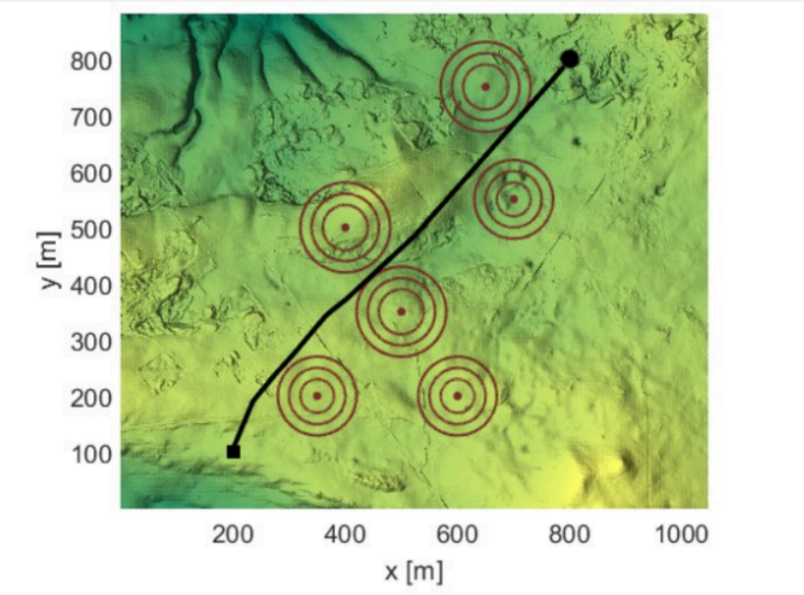
- Minimize Total Cost
- Lower cost → Better UAV path

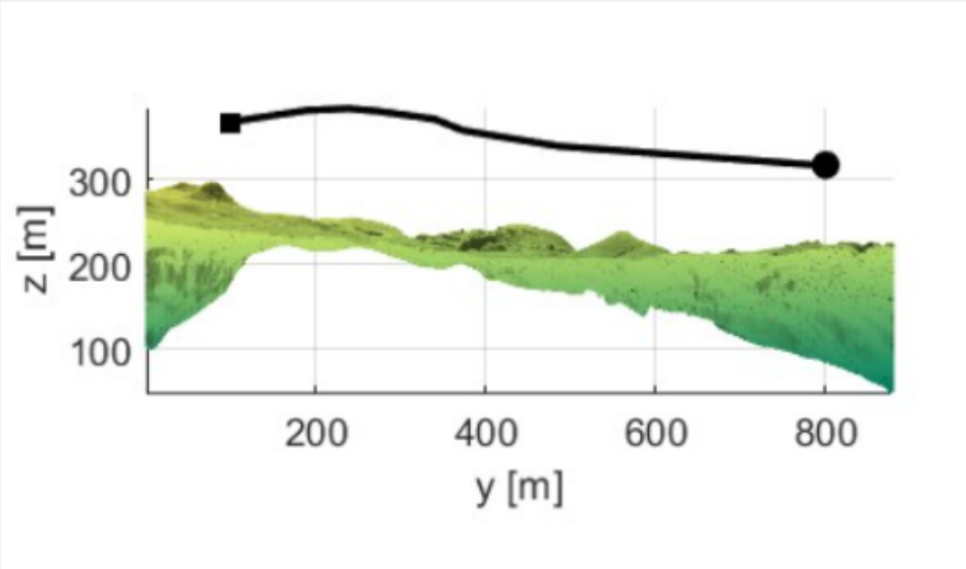


3D UAV Path Planning Result

Optimal UAV Path in Terrain Environment

- Terrain map with elevation
- Cylindrical obstacles represent threats
- Black line shows optimized UAV trajectory
- Safe path avoiding obstacles

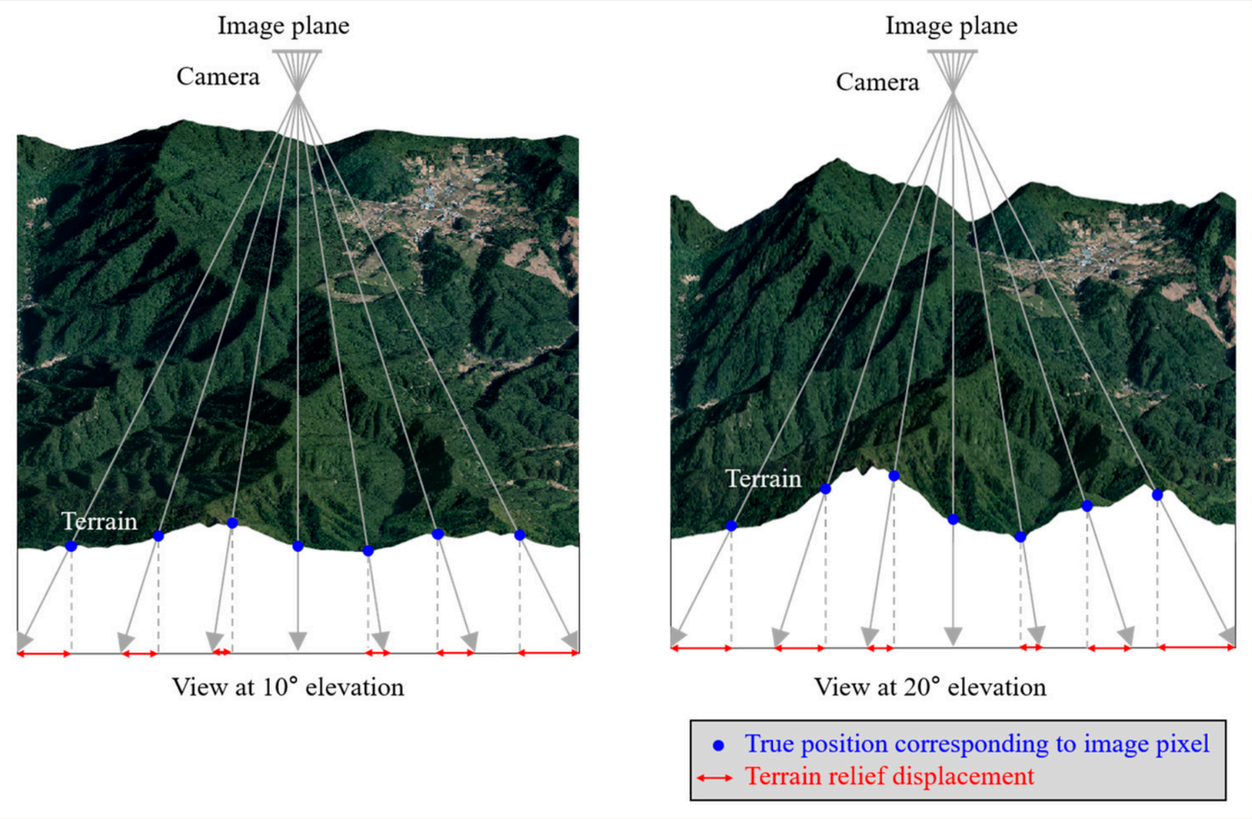




Side View of UAV Path

Altitude Profile Analysis

- Side view of UAV path
- Terrain elevation shown
- UAV maintains safe altitude
- Smooth trajectory generation



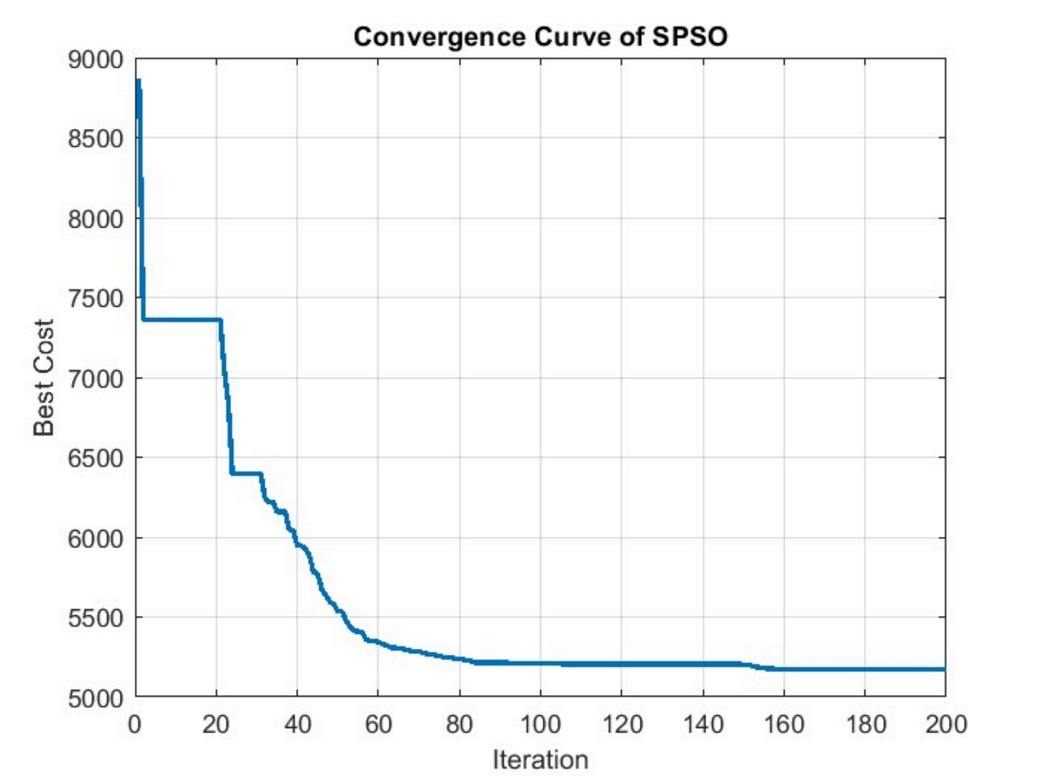
Convergence Curve

Convergence Analysis

- X-axis → Iterations
- Y-axis → Best Cost Value

Observations

- Cost decreases gradually
- Algorithm improves solutions over time
- Converges to stable optimal value

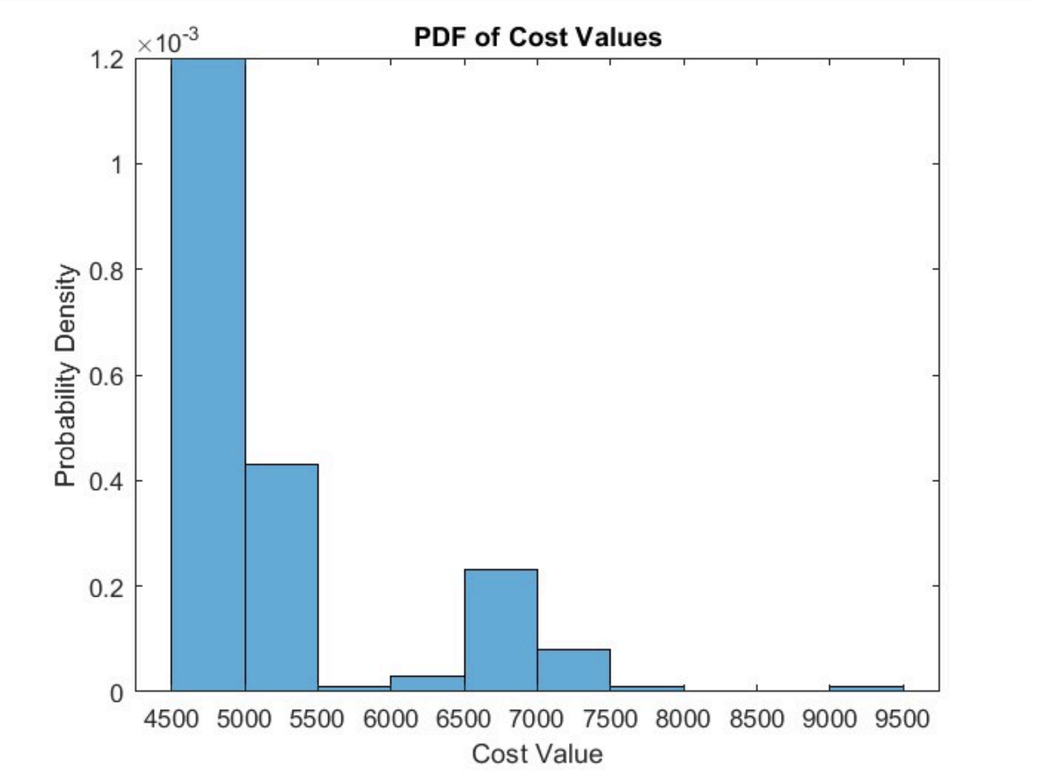


Probability Analysis (PDF & CDF)

Statistical Analysis

Probability Density Function (PDF)

- Shows distribution of final path cost values
- Indicates how frequently each cost occurs
- Concentration near lower cost → better solutions



Probability Analysis (PDF & CDF)

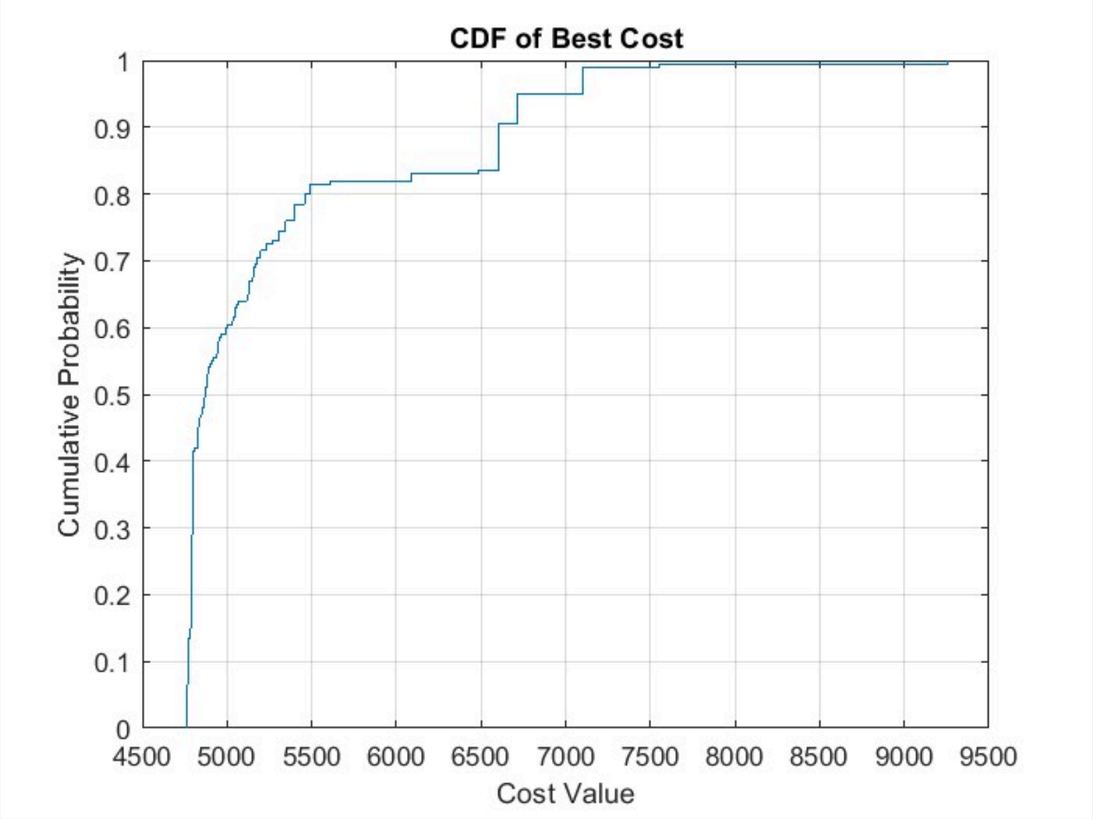
Statistical Analysis

Cumulative Distribution Function (CDF)

- Shows probability that cost ≤ given value
- Helps evaluate reliability of the algorithm
- Steeper curve → higher chance of good solutions

Observation

- High probability of low-cost solutions



Conclusion

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- SPSO effectively solves UAV path planning
- Generates safe collision-free trajectories
- Handles complex terrain environments
- Convergence analysis shows algorithm stability
- Statistical analysis confirms reliability

Future Work

- Dynamic obstacle avoidance
- Real-time UAV navigation